Project Design Phase

Solution Architecture

Date	8 September 2026
Team ID	Individual
Project Name	Automated Incident Management Lifecycle using ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for the Automated Incident Management Lifecycle project bridges the IT incident-management problem with a ServiceNow-based technology solution.

Its goals are to:

- Provide a standardized process for creating, classifying, assigning, and tracking incidents.
- Define how incident data moves from creation and classification through knowledge assistance, escalation, and resolution.
- Integrate Level 2 support, emergency change requests, child incidents, Knowledge Management, and SLA tracking.
- Provide a clear architecture that can be configured, tested, maintained, and extended within ServiceNow.

Project-specific architecture:

An end user reports an IT issue and the Service Desk creates an incident in Service Operations Workspace. The incident is classified using Category, Subcategory, Urgency, Service, Service Offering, and Configuration Item. Agent Assist retrieves relevant Knowledge Articles. The incident can be assigned or reassigned to Level 2 support. When required, an emergency change request or child incident is created from the incident. Level 2 technicians investigate, document probable cause and resolution, and resolve the incident. A Knowledge Article can be created from the resolved incident, while SLA and related records provide service visibility.

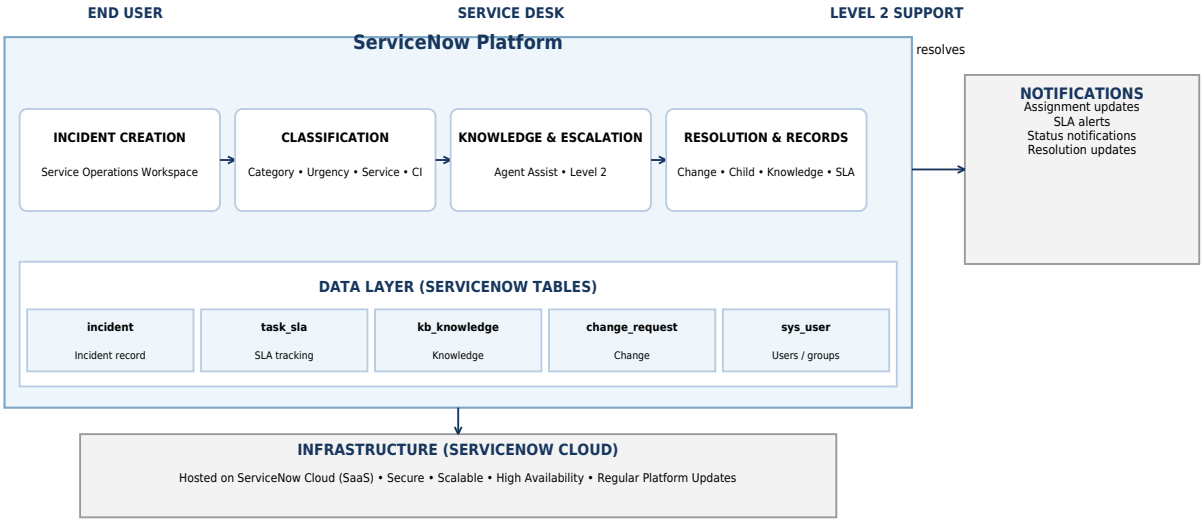


Figure 1: Solution architecture and data flow of the ServiceNow Automated Incident Management Lifecycle

Architecture Components:

Component	Purpose
End User / Caller	Reports an IT issue and provides the initial problem information.
Service Desk / Service Operations Workspace	Creates the incident, captures caller and short description, and manages initial classification and assignment.
Incident Management	Stores and manages the incident lifecycle, including category, urgency, service, configuration item, state, and activity.
Agent Assist / Knowledge Management	Displays relevant Knowledge Articles and allows useful articles to be attached to the incident.
Level 2 Support	Investigates complex incidents, updates work notes, changes configuration items when required, and resolves the issue.
Change Management	Supports creation and linkage of emergency change requests when incident resolution requires a change.
Child Incident	Tracks recurring or related issues as separate child records linked to the parent incident.
Resolution & Knowledge	Captures probable cause, resolution code, and notes, and supports Knowledge Article creation from resolved incidents.
SLA & Related Records	Tracks SLA progress and provides visibility of child incidents, changes, knowledge, and other related records.
ServiceNow Platform / Update Sets	Hosts the configured workflow and supports controlled configuration management and migration between instances.